

Computational screening of sustainable low-emissivity PVD coatings on soda-lime float glass

Technical report · Saint-Gobain Research India

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Framework: [pvdlowe](#) v0.1.0 — <https://github.com/VarunMandy/pvdlowe>

Abstract

A computational framework was built to execute the screening workflow set out in the project brief: element screening, thermodynamic stability, optical multilayer modelling, design of experiments and multi-objective ranking, for cost-effective low-emissivity coatings deposited by argon magnetron sputtering.

Applying it produced three classes of result. **First**, the proposed multi-objective weighting was found to contain six defects, one of which allowed a thickness optimiser to return a design consuming 29% *more* silver than the incumbent while scoring 100/100. **Second**, two of the brief's literature citations could not be matched to any locatable source, one transcribed value was traced to a conflict between two publications by the same group, and one pivotal measurement was found inconsistent with a model-independent electrodynamic limit, with six explanations tested and eliminated. **Third**, the design-space search identified a material class and a stack architecture outside the brief's framing, both of which outperform the proposed candidates: silicon nitride dielectrics improve visible transmittance by up to nine percentage points over AZO, and the composition optimum lies at 5–15 at.% Ag rather than the proposed 70%.

Thermodynamic screening of all fifteen nominated systems confirmed that Ag-Cu possesses no stable ordered compound, supporting the brief's suspicion of phase separation with first-principles evidence. A machine-learning interatomic potential extended this to the disordered solid solution and showed that in the dilute-silver range the driving force to separate falls below thermal energy at deposition, supplying a third independent argument for that composition range. A literature search subsequently identified the mechanism behind the dielectric effect — underlayer templating of metal grain structure, measured by transmission electron microscopy in the patent literature — and established that it is the same physical effect as the framework's largest known error.

No films were deposited. All performance figures are model outputs. The framework's own validation identifies its largest error — an approximately eightfold under-prediction of sputtered copper sheet resistance — in the material on which most leading candidates depend. A specified experiment resolving this, with pre-registered decision rules, is delivered with the framework.

1. Scope and deliverables

1.1 What was asked

The brief proposed a materials-by-design programme for transparent low-emissivity coatings on float glass, centred on a dielectric/metal/dielectric architecture (AZO/metal/AZO) and asking how much silver could be removed while retaining visible transmittance, sheet resistance and far-infrared reflectance.

1.2 What was delivered

Deliverable	Status
Framework implementing the brief's §21 workflow	Complete — 9,586 lines, 136 tests
Element and compound screening	Complete
Optical multilayer model (transfer matrix, EN 410 / 673 / 12898)	Complete, validated
Thin-film transport coupled to optics	Complete, uncalibrated — requires §6 experiment
Multi-objective scoring with provenance grading	Complete
Design of experiments generator	Complete
DFT input generation (VASP)	Complete, not executed — requires licence and HPC
Materials Project integration	Complete and executed — 15/15 systems
ML interatomic potential surrogate	Complete and executed — mixing energies
Candidate ranking, 38 architectures	Complete, model outputs only
Predicted XRD signatures for the experiment	Complete
Experimental protocol, pre-registered analysis	Complete, not executed — requires tool access

Repository, documentation, results and protocols are version-controlled and archived.

1.3 How to read this report

§2-§3 do not depend on any experiment. They are audits of the brief's method and evidence, and they stand on their own.

§4-§5 are model outputs. They are hypotheses ordered by physics, suitable for directing experimental effort, and their principal weakness is quantified in §6.

§7 is what only Saint-Gobain can answer.

2. Method audit: six defects in the multi-objective weighting

The brief's §14 specifies criteria and weights. Encoding that table literally produced a scoring function that did not measure the project's stated objective. Four defects were in the table as proposed; **two more were found by re-auditing the corrections for the first two**. All were surfaced by the framework's own diagnostics ([pvdlowe check-weights](#)).

2.1 Silver consumption carried zero weight

§14's table has no line for silver mass, although §17 and §20 both name minimising silver as a project objective. Silver mass was computed, displayed, even reported as the limiting criterion — and contributed nothing to the score.

The thickness optimiser exploited this immediately, returning a design using **29% more silver than the benchmark, at a perfect 100/100**.

Correction: silver mass weighted 0.15. See §2.7 on why that number is itself a judgement.

2.2 Emissivity and sheet resistance double-count one property

They correlate at **$r = 0.978$** across the candidate set. Both are the free-carrier response of the same layer — the same physics measured two ways — so weights of 0.25 and 0.15 placed 0.40 of the total on one quantity.

Correction: sheet resistance weighted 0.0 and reported as a constraint.

2.3 Targets set at the specification minimum, causing saturation

Derringer-Suich desirability saturates at 1.0 once a target is reached. Setting T_{vis}'s target to 0.80 — the brief's *floor of acceptability* — made the criterion flat exactly where candidates differ.

Across 181 oxide pairs clearing 0.80, transmittance spanned 0.800–0.881 while the **score went down**, 74.99 to 73.30.

Correction: targets raised to aspirational values. **General rule: a desirability target should be an aspiration, not a specification minimum.**

2.4 The supply-risk band could not discriminate

With floor 8.0 against candidate values of 4.5–7.5, all silver-bearing compositions scored 0.1–0.28, and supply risk was reported as limiting for 12 of 14 candidates.

Correction: floor 9.5, target 4.0 — still zeroing indium-bearing ITO, which is the criterion's purpose.

2.5 A triple-count the fix for §2.2 left untouched

Zeroing sheet resistance removed one double-count and left another:

Pair	r
Silver mass ↔ metal cost	1.000
Silver mass ↔ supply risk	0.991

Silver dominates metal cost so completely that cost is the same quantity in different units, and supply risk in this candidate set is silver's supply risk. Together they carried **36% of the effective weight** on one property.

Correction: both weighted 0.0 and reported as derived. Effective concentration falls to 25%, and no criterion pair now both carries weight and correlates.

The diagnostic itself was also wrong — it flagged correlated pairs regardless of whether both carried weight, which buries the cases that matter. It now distinguishes an active double-count from an informational correlation.

2.6 Weight reserved for criteria never populated

Structural stability, thermal stability and deposition efficiency carried **0.30 of a 0.90 nominal total while being None for every candidate**. The scheme renormalises over available criteria, so that third was silently redistributed: a file stating emissivity at 0.25 was in fact applying 0.42.

Worse, `structural_stability` carried 0.15 **with no criterion definition at all** — it could never have scored even had a value been supplied.

Correction: all three weighted 0.0, the missing definition written. **The ranking is unchanged, which is the point** — they were renormalising away already.

`structural_stability` is now partly populatable from the Materials Project screening of \$5.5 and the surrogate mixing energies of \$5.6, both in eV/atom. It is left at 0.0 deliberately: populating it would change every ranking, the values carry a 20–40% systematic error, and 0.15 is another weight nobody derived. **Populating a criterion and choosing its weight are one decision, not two**, and that decision belongs to whoever continues the work.

2.7 The ranking depends on a number nobody derived

Silver mass carries a weight of 0.15. That value is a judgement, and it selects the answer:

Silver weight	Winner	Ag g/m²	Runner-up	Margin
0.00	N35e	0.065	N3e	0.52
0.05	N35e	0.065	E10e	0.09
0.10	E10e	0.015	E5e	0.57
0.15	E10e	0.015	E5e	0.22
0.20	E5e	0.008	E10e	0.09
0.25	E5e	0.008	N6	0.15
0.30	N6	zero	E5e	0.17
0.40	N6	zero	E5e	0.70

The winner changes three times across a plausible range, and five of eight settings separate first from second by under half a point — at those weights the ranking is not meaningfully distinguishing candidates.

So the honest statement is the sweep, not the ranking: the recommendation is E10e at weight 0.15, E5e at 0.20–0.25, and the silver-free N6 at 0.30 and above. Choosing among them is a decision about how much silver consumption matters to Saint-Gobain, and it is not the framework's to make. Run `pvdlowe check-weights` to reproduce it.

2.8 Aggregation

The brief states its weighting should prevent a candidate "winning simply because it has excellent conductivity while being unacceptable optically." A weighted arithmetic sum does not achieve this — a candidate scoring zero on one criterion forfeits only that criterion's weight. **A geometric mean sends the whole score to zero.**

The framework uses geometric aggregation by default and can report both, since the ordering differs materially between them.

3. Evidence audit

3.1 Verification status

Every citation in the brief carried `utm_source=chatgpt.com`, meaning the sources were surfaced by a language model rather than read from a publisher. That does not make them wrong — several are reproduced well by the model, which is weak evidence they are right — but none had been checked.

Status	Count
Verified — full text read	3
Partial — source located, figures confirmed against abstract	4
Disputed — figures appear in no locatable source	2
Supported — corroborated independently	3
Not located after searching	1

3.2 Two citations cannot be supported

The figures 85.4% T_{vis} / 3.21 Ω/sq / 97% FIR, and 78.7% T_{vis} / 2.7 Ω/sq , appear in neither located AZO/Ag/AZO study. **They should not be cited until someone establishes where they came from.**

An uncomfortable corollary. These two entries are the model's three *closest* agreements — 0.3%, 1.0% and 1.2% relative error. Excluding them, the median validation error rises from **14.7% to 30.6%**. The framework's apparent accuracy rested partly on figures that cannot be traced.

`validate_model()` now excludes them by default and reports 30.6%; 14.7% is available only behind a flag, labelled as an audit view that must not be quoted. A test fails if the exclusion ever starts *lowering* the reported error.

3.3 A measurement inconsistent with a model-independent limit

Applied Surface Science **578** (2022) reports AZO/Cu/AZO with T_{vis} 87.7%, R_s 9.96 Ω/sq and ϵ 0.055. **The brief's §16 conclusion — that a silver-free stack may already meet a Low-E target — rests entirely on this measurement.**

For a conducting sheet much thinner than the wavelength, far-infrared reflectance is fixed by sheet resistance alone through the impedance of free space. No fitted parameter enters. At 9.96 Ω/sq the floor is $\epsilon \geq \mathbf{0.096}$; the reported 0.055 would require about 5.48 Ω/sq .

Six explanations were tested and eliminated:

Explanation	Outcome
Different samples	Abstract attributes all three figures to one sample
Transcription error	Confirmed against the abstract
Far-IR glass index assumption	Limit is 0.091–0.097 for $n = 1.5\text{--}4.0$
Band-limited emissometer	Reads 5–8% higher — moves it the wrong way
This framework's own convention	An industrial formula in use since 2000 agrees within 10%
Measurement basis	A hemispherical reading fails by 51% rather than 42%

The last of these was the final benign explanation. The impedance floor is a normal-incidence quantity, so the open question was whether 0.055 might be hemispherical and therefore not comparable. Reading it that way gives a normal-basis equivalent of 0.0472 — **worse, not better**.

§16's conclusion should not be used until this is settled.

3.4 The two copper measurements are in opposite positions

Both copper benchmarks, checked against the same bound:

	Group, technique	R_s	ϵ	Floor	
Miao 2014, HK PolyU	RF	16.60	0.330	0.150	consistent
ApSS 2022, Xi'an	DC	9.96	0.055	0.096	below floor

The measurement that obeys the electrodynamic limit is the one that contradicts this framework, by a factor of 7.7 — the model gives 2.16 Ω/sq for a 15 nm copper layer against 16.6 measured. The measurement that agrees with the model's optimism is the impossible one.

That contrast is stronger evidence than either entry alone, and it points the same way as §6: the fault is in the model, not the literature.

They must not be pooled — different groups, RF against DC, different decades, and the 2014 study is from a textiles laboratory

that coats polyester fabric with the same stacks.

3.5 Corrections to the brief's characterisations

- **Deposition methods differ across the benchmark set.** The AZO/Ag/AZO trilayers used **RF** sputtering where §3 implies DC; the AZO single layer used **medium-frequency**. Film density and resistivity do not transfer between techniques, so these sources should not be pooled.
- **One transcribed resistivity is a source conflict, not a transcription error.** The brief quotes 1.59 $\mu\Omega\cdot\text{cm}$ for a 10 nm Ag film. The journal states 1.59 — the brief is faithful — while the group's own patent states 1.29, which is *below* bulk silver and therefore impossible. **Both figures are anomalous by a factor of three:** Fuchs-Sondheimer gives 4.74 $\mu\Omega\cdot\text{cm}$ at 10 nm, this framework gives 4.44, and an independent 12 nm measurement gives 4.8. Cite the comparative Ag-Cu against Cu claim; do not cite 1.59 as a validated thin-film benchmark.
- **The Ag₇₀Cu₃₀ basis is on the wrong substrate.** The brief's §5 justification traces to a 2022 paper that does support 70/30 at 10 nm — but on **polycarbonate**, for first-surface coatings. Given §5.2, that optimum does not transfer to a glass line unchecked.

4. The specification, completed

The brief supplies three constraints. Two more matter and were unspecified, so they have been anchored to the Indian building code rather than invented.

4.1 Against the brief's three constraints

Criterion	Spec	Pass	Achieved range
T_vis	≥ 0.80	31/38	0.738 – 0.918
R_sheet	$\leq 5.0\ \Omega/\text{sq}$	19/38	2.63 – 7.55
ϵ_h	≤ 0.10	38/38	0.042 – 0.098

All three simultaneously: **15 of 38**.

Emissivity never binds. The brief's target is generous relative to what any percolating metal layer gives automatically. Sheet resistance is the real constraint, then transmittance.

4.2 Two criteria anchored to ECBC 2017

Criterion	Anchor	Value
g_value	ECBC prescriptive SHGC, composite climate	≤ 0.27
U_g	ECBC assembly U-factor	$\leq 3.0\ \text{W/m}^2\text{K}$

Basis warning. ECBC specifies *assembly* values — frame and glass, area-weighted. This framework computes centre-of-glass, and its g-value is single-pane with the standard $N = 0.36$ inward fraction. The conversion must be stated rather than assumed.

Chaining a clear second pane ($T_{\text{sol}}\ 0.83$) gives an approximate DGU SHGC:

	g single	g DGU	vs 0.27
M6, the best	0.558	0.463	fails by +0.19
N35e, the worst	0.793	0.658	fails by +0.39

Not one of the 38 candidates meets the Indian code's solar-heat-gain requirement, and the best misses by seventy per cent.

That is external, regulatory confirmation of §5.4: these are thermal Low-E coatings, not solar-control coatings. For Indian commercial glazing the single-metal architecture is not merely suboptimal — it is non-compliant, and the double-metal stack of §5.4 becomes a requirement rather than a refinement.

U-value is not binding. Every candidate lands at 1.12–1.29 $\text{W/m}^2\text{K}$ centre-pane; even after a frame adds 0.5–1.5 there is room.

4.3 The two targets are mutually incompatible

Searching 990 double-metal designs against all five constraints returns **none**. The frontier:

If T_vis at least	Lowest g reachable	LSG
0.80	0.424	1.89
0.70	0.351	2.04
0.65	0.326	2.08
0.60	0.271	2.26

ECBC's 0.27 is reachable — at T_vis 0.644. The brief asks for 0.80.

Since ECBC's own visible-light minimum is 0.27, the incompatibility is not with the regulation but with the brief's **self-imposed** transmittance target, which was written for a heating climate. **Which target is real is a decision for Saint-Gobain.**

5. Design-space results

5.1 The composition optimum lies at 5-15 at.% Ag

Eight composition curves — Ag 0-100% in 5% steps, geometry re-optimised at every point, across two microstructure models, two dielectrics and two climates.

Every one peaks between 0 and 10 at.% Ag.

Profile	Microstructure	Dielectric	Optimum	Plateau
Heating	Segregated	Si ₃ N ₄	0.05	0.00-0.20
Heating	Segregated	AZO	0.10	0.05-0.40
Cooling	Either	Either	0.00	0.00-0.15

This is a sustainability result, not a performance one. Emissivity and sheet resistance keep improving up to 50 at.% Ag. The score falls anyway because silver mass carries weight: the trade is about 0.005 in emissivity for an 87% cut in silver, and it is the right trade only because silver was decided to matter.

Plateau width matters more than peak position for a production process.

5.2 The dielectric changes the metal, not just the interference stack

The framework originally concluded that Si₃N₄ beats AZO on transmittance by up to nine points, on index matching. **A measurement reversed it.**

Cueva & Carretero (*Coatings* **13**, 1709, 2023, open access, **full text read**) deposited five dielectrics under identical conditions with 10 nm Ag:

Dielectric	Measured ε	n(550 nm)
SnO ₂	0.083	2.00
ZnO	0.064	2.019
AZO	0.058	1.85
SiAlN _x	0.067	2.09

AZO wins on emissivity and transmittance despite the lowest refractive index of the five. The reason the authors give: silver grows more efficiently on AZO.

What the model was missing. It treated the dielectric purely as an interference layer — index and thickness, nothing else. The optical inputs were sound; their measured indices match the framework's closely. What was absent is that the underlayer determines the *quality of the metal grown on it*.

Correction: [metal_growth_factor](#) per underlayer, calibrated to that series and normalised to AZO = 1.00: ZnO 1.10, SiAlN_x 1.16, ITO 1.20, TiO₂ 1.25, SnO₂ 1.43. The model now reproduces the measured ordering and magnitudes to 4-8%.

The productive outcome is a hybrid, which is what industry already does:

ID	Stack	T_vis	ε_h	Heating score
H1	AZO/Ag/Si ₃ N ₄	0.918	0.057	69.6
M0	AZO/Ag/AZO (benchmark)	0.876	0.060	66.6
H2	AZO/Cu/Si ₃ N ₄ (silver-free)	0.788	0.042	77.6

H2 is the only architecture appearing in both climate top-tens.

5.3 Climate reverses the ranking

The brief does not specify a climate, and its weighting does not constrain solar heat gain — which quietly encodes a heating-dominated assumption.

Adding the solar-gain metrics and re-scoring, the two top-ten lists **share one candidate out of ten**.

Heating-dominated — [data/targets.yaml](#)

#	ID	Architecture	T_vis	ε_h	R_s	Ag g/m²	Score
1	E10e	Si3N4/Ag10Cu90/Si3N4 (segregated)	0.877	0.051	3.13	0.015	87.4
2	E5e	Si3N4/Ag5Cu95/Si3N4 (segregated)	0.863	0.048	2.98	0.008	87.1
3	N6	Si3N4/Cu/Si3N4	0.858	0.054	3.41	0.000	86.2
4	E5	Si3N4/Ag5Cu95/Si3N4	0.861	0.061	3.99	0.008	83.7
5	N35e	Si3N4/Ag60Cu40/Si3N4 (segregated)	0.903	0.050	3.04	0.065	81.5
—	M0	AZO/Ag/AZO (benchmark)	0.876	0.060	4.18	0.105	66.6

Cooling-dominated (India) — [data/targets_cooling.yaml](#)

#	ID	Architecture	T_vis	g	LSG	Ag g/m²	Score
1	M6	AZO/Cu/AZO	0.738	0.558	1.32	0.000	59.8
2	H2	AZO/Cu/Si3N4 (hybrid)	0.788	0.631	1.25	0.000	55.8
3	S10	ITO/Ag70Cu30/ITO	0.848	0.619	1.37	0.081	53.9
4	M3ema	AZO/Ag70Cu30/AZO (segregated)	0.840	0.635	1.32	0.081	53.7
—	M0	AZO/Ag/AZO (benchmark)	0.876	0.646	1.35	0.105	50.6

How to read these tables:

- 1. **They are not a ranking until the silver weight is fixed** — see §2.7.
- 2. **They share one candidate out of ten**, H2.
- 3. **The benchmark appears in neither top ten**.
- 4. **All copper-bearing entries are contingent on \$6**.
- 5. **Scores are not comparable between the two tables** — different weighting files, different criteria.
- 6. **No cooling-profile entry meets the transmittance specification**.

The mechanism is physical. AZO is a transparent conductive oxide: free carriers put a screened plasma edge at 1.25 μm, inside the solar band. Si3N4 is a passive insulator. Same 11 nm copper layer, transmittance at 1200/1600/2000 nm: AZO 0.240/0.115/0.068 against nitride 0.509/0.259/0.151.

5.4 Single-metal architectures cannot achieve solar control

The best light-to-solar-gain ratio across all 38 single-metal candidates is **1.37**, against roughly 2.0 for commercial solar-control glazing.

Architecture	Metal	Dielectric	Geometry (nm)	T_vis	g	LSG	ε_h	R_s
Single	Cu	AZO	35/12/35	0.738	0.558	1.32	0.046	2.87
Double	Ag	Si3N4	15/12/60/12/15	0.819	0.464	1.76	0.025	1.67
Double	Cu	Si3N4	59/13/105/13/59	0.716	0.516	1.39	0.024	1.33

LSG 1.76 against a ceiling of 1.37 — 28% better. A single film trades transmittance against infrared reflectance along one curve; the dielectric *between* two metals adds an interference degree of freedom, so the two reflections can cancel in the visible while adding in the infrared.

Note the optimum geometry: **thin outer dielectrics, thick middle**. A symmetric guess misses it.

A third layer is not worth adding. LSG goes 1.33 → 1.76 → 1.81 for n = 1, 2, 3 — the third buys 0.05 for a 25% silver increase, and for copper it collapses transmittance to 0.437.

But silver consumption doubles, since each layer must independently clear percolation near 10 nm, so double-silver scores badly on sustainability despite being the best solar-control design found.

5.5 Thermodynamic screening

All fifteen nominated systems retrieved from the Materials Project and screened within 50 meV/atom of the convex hull.

Two intermediate Ag–Cu phases exist and **neither is stable**: Cu₃Ag at **0.0904 eV/atom** above the hull, CuAg₃ at **0.0857**. Ag–Cu is the only nominated binary with no stable intermediate phase, and 86–90 meV/atom is well above k_BT at deposition.

A second finding, bearing on the ternary hypothesis. Cu–Ti has 8 near-hull phases and Al–Cu has 10. A dilute Ti or Al addition therefore has stable intermetallics available to precipitate rather than remaining in solution as a stabiliser — a second, independent mechanism for the ternary underperforming, alongside the Nordheim conductivity penalty.

Conclusion. The brief's §5 suspicion of phase separation, inferred from a 13% lattice mismatch, is now supported by first-principles data.

5.6 Mixing energies: the dilute corner is thermodynamically favoured

§5.5 is an equilibrium statement about *ordered* phases. A sputtered film is neither ordered nor at equilibrium, so the disordered solid solution needed separate treatment.

MACE-MP-0 on 32-site fcc supercells, gated against the two known hull distances before any prediction was accepted. Positive across the whole range, fitting a regular-solution form to ±6 meV/atom: **ΔE_{mix} = 0.287·x·(1–x)**.

A self-check that matters more than the gate. At Ag 25 at.% the surrogate puts the *disordered* solution 13 meV/atom **below** the ordered Cu₃Ag compound — no ordering tendency at all, which is precisely what an empty hull implies, reached independently.

Ag at.%	ΔE _{mix}	×kT (550 K)
70 — the brief's priority	0.0602	1.27
15	0.0366	0.77
10	0.0258	0.54
5	0.0136	0.29

In the dilute-silver corner the driving force to separate falls below thermal energy at deposition. §5.1 found the two microstructure hypotheses converge there and treated it as robustness; the thermodynamics supply the reason.

The dilute optimum is therefore supported three independent ways: lowest silver consumption, highest corrected score, weakest tendency to phase-separate.

Cross-checked. CHGNet reproduces the shape; both models under-predict — MACE by 0.026, CHGNet by 0.047 eV/atom — so the bias is inherited from shared training data rather than architecture-specific, meaning ΔE_{mix} should be corrected upward. **They are not independent:** both train on the same Materials Project trajectories, so agreement shows the bias is consistent, not absent.

A compounded correction. Verifying the brief's own claim 7 established that GGA underestimates Cu-transition-metal formation energies by nearly 40%, because it places Cu-3d bands too shallow. The MP hull is GGA/GGA+U and both surrogates train on it, so two errors act in series. The conclusion **holds at 5–10 at.% Ag and becomes marginal at 15%.**

What a surrogate cannot do. Optical constants, dielectric functions, band gaps, emissivity — these models carry no electronic structure. Every optical number here comes from the transfer-matrix model or from measurement. And a conductivity result does not transfer to an optical one: the emissivity-sheet resistance link assumes a low-damping Drude metal, a property of *crystalline* silver.

5.7 The nucleation mechanism, and a structural weakness it exposes

§5.2 established that the dielectric changes the metal's growth and encoded it empirically. Two surrogate calculations were run to find the mechanism. **Both failed:**

Attempt	Failure mode
Bulk interface adhesion	5–21% lattice mismatch — measured elastic strain, not binding
Adatom wetting	ZnO(0001) termination changed the result by 2.1 eV, against 0.46 eV between materials

A literature search resolved it in under an hour. US 7,632,572 B2 / US 8,512,883 B2 (AFG Industries → AGC → Cardinal CG; **full text read**) deposited 16 nm Ag on amorphous TiOx and on a 5 nm ZnO seed over the same TiOx, and examined both by TEM:

- **25 nm grains on ZnO against 15 nm on a-TiOx**
- {111}-oriented grains **two to three times larger** on ZnO
- the film on bare amorphous titania **clearly discontinuous** where the ZnO-seeded film was continuous across the whole specimen

The inventors name the mechanism. Zinc oxide grows {0001}, which orients the silver to grow {111}, and the epitaxial lattice match between Ag{111} and ZnO{0001} lowers sheet resistance and improves adhesion.

And it validates this framework quantitatively. The patent reports four-point sheet resistance — **5.68 Ω/□ with the ZnO**

seed, 7.56 without, a ratio of **1.331**. The `metal_growth_factor` values of \$5.2 were calibrated independently from a different group's *emissivity* series and give **1.250**.

	Ratio, titania-like : zinc-oxide-like
Patent, four-point sheet resistance, 16 nm Ag	1.331
This framework, calibrated to emissivity	1.250
Agreement	93.9%

This is the first independent quantitative validation of a parameter in this work: fitted to one dataset, it reproduces another it never saw. The comparison is not exact — the patent contrasts a ZnO seed *over* titania against titania alone — so it is corroboration rather than confirmation.

A percolation datum follows. The patent claims continuous, strongly adherent silver **down to 8 nm** on a zinc oxide seed, against the 10 nm this framework assumes. A 20% difference, and one that would move every silver-consumption figure in \$5.1.

The structural weakness. The framework models each metal layer as though the layer beneath it did not shape its microstructure. That single omission accounts for **both** `metal_growth_factor` **and** the eightfold copper under-prediction of \$6. They are one effect appearing twice, not two separate caveats.

A successor should replace the empirical multiplier with a measured `grain_size_ratio` per underlayer, which is physically meaningful, directly measurable, and feeds both the optical and electrical paths through the existing Mayadas-Shatzkes term.

6. The critical unknown, and the experiment that resolves it

6.1 What is wrong

The framework validates well against silver in the infrared — within 0.4% on the benchmark it reproduces best — and **fails on copper**, under-predicting sputtered sheet resistance by roughly eightfold. **Six of the leading candidates in both climate profiles are copper-based.**

The classical size effect is eliminated. Four independent published fits give speculariry 0.48–0.80 and grain-boundary reflection 0.16–0.43. This framework's assumed 0.50 and 0.25 sit inside that range, and an exhaustive scan over the entire admissible range reaches only **4.42 Ω/sq against the 16.6 to be explained** — short by a factor of 3.8.

Grain structure is the leading hypothesis: grains at roughly half the film thickness give 14.1 Ω/sq.

And a seed layer recovers most of the gap. Six further AZO/Cu/AZO studies were located:

Film	Seed	R _s
AZO/Ti/Cu/AZO, <i>Opt. Lett.</i> 2017	1–2 nm Ti	4.31 Ω/sq
AZO/Cu/AZO, <i>ApSS</i> 2022	none	9.96
AZO/Cu/AZO, <i>Miao</i> 2014	none	16.6
this framework, 15 nm Cu	—	2.16

The framework is eightfold optimistic against unseeded copper and roughly **twofold** against seeded copper. **So the model is not wrong about copper as a material — it is wrong about poorly nucleated copper**, which is the same mechanism as \$5.7 in a third independent place.

If that reading is right, copper's obstacle is film quality, not physics — an engineering problem with known levers rather than a fundamental limit.

6.2 The experiment

One XRD scan, one film, an afternoon — and it answers three questions.

Measurement	Question answered
Peak count and position	Microstructure. Segregated gives two fcc sets; a solid solution gives one at a Vegard-interpolated angle
Scherrer peak width	Grain size , and therefore <code>grain_size_ratio</code> — the parameter \$5.7 identifies as the common cause
(111)/(200) ratio vs the powder value of 2.2	Templating , the mechanism of \$5.7

`pvdlowe.characterise` states in advance what the scan should show under each hypothesis, so it can be planned rather than interpreted afterwards.

A second, independent discriminator. Sheet resistance separates the microstructure hypotheses by a number — 6.4 against 2.6 Ω/sq . Diffraction separates them by *peak count*. Two unrelated observables agreeing is far stronger than either alone, and both come from one film.

Four requirements, all cheap:

- 1. **Scan to 20 = 100°, not 60.** Separation grows faster than width with angle.
- 2. **Grazing incidence, not Bragg-Brentano.** A 2010 study of this exact architecture reports *no detectable Cu peaks* for 3–13 nm layers, while ApSS 2022 sees a weak (111) at 43.8° — matching this framework's predicted 43.32°. A copper peak here is **marginal**, and a null would otherwise be ambiguous between "nanocrystalline" and "insufficient signal".
- 3. **A 30 nm uncapped calibration film**, to prove the measurement works before applying it to the marginal case.
- 4. **A seeded arm** — one extra target, per \$6.1.

Pre-registered decision rules are in `experiments/PROTOCOL_cu_series.md` — four outcomes, each with what it means and what to do, **including the one where the literature is right and six candidates fall over**.

7. The question only Saint-Gobain can answer

This report benchmarks against a 10 nm AZO/Ag/AZO at ϵ_h 0.060, taken from the brief — **whose citations for that stack are the two disputed entries of §3.2**.

Saint-Gobain's own patents (US 7,745,009) claim silver at **12.5–16 nm targeting $\epsilon \leq 0.038$** . This model says that needs 14–16 nm.

Ag nm	ϵ_h	Ag g/m ²
10.0 (current benchmark)	0.060	0.105
14.0	0.039	0.147
16.0	0.033	0.168

If production runs at 12.5–16 nm, the silver reductions reported here are measured against a stack that would not meet Saint-Gobain's own emissivity specification. The baseline needs confirming before the percentages are quoted.

Alongside it, §4.3: **which transmittance target is real** — the brief's 0.80, or ECBC compliance at 0.644. They cannot both be met.

8. Limitations

8.1 No experimental validation

Every performance figure is a model output. Median error against traceable literature is 30.6%.

8.2 The principal structural weakness

The framework models each metal layer as though the layer beneath it did not shape its microstructure. §5.7 establishes that this single omission accounts for both the empirical `metal_growth_factor` and the eightfold copper under-prediction of §6. Grain size in a sputtered film is set substantially by the underlayer — 25 nm on crystalline ZnO against 15 nm on an amorphous oxide, measured by TEM — and grain size feeds directly into the Mayadas-Shatzkes term the framework already contains.

8.3 Calibrated and fitted parameters

Specularity and grain-boundary reflection are fitted, not measured. Si_3N_4 and TiO_2 optical constants are `ESTIMATE` grade. `metal_growth_factor` is calibrated to one measured series, with copper untested — though §5.7 supplies independent corroboration to 6%.

8.4 Illuminant approximation, quantified

Varying colour temperature over 5000–7500 K moves **T_{vis} by 0.1%** and **T_{sol} by 8.3%**. T_{vis} is reportable as computed; T_{sol}, g-value and LSG carry a 5–10% systematic until tabulated AM1.5G spectra are supplied.

8.5 Scores are not portable between weightings

Absolute scores moved by roughly 25 points as the §2 defects were fixed. **A score is not a property of a material.** Always

quote the weighting file.

8.6 Not modelled at all

Roughness, interdiffusion, damp-heat and abrasion durability, adhesion, agglomeration kinetics, amorphous metal layers, and any property of a metastable sputtered alloy that a 0 K ordered-crystal database cannot supply.

9. Conclusion

What this work established. A better composition — 5–15 at.% Ag rather than 70%, supported on three independent grounds. A better material class — silicon nitride, outside the brief's search framing, with the hybrid AZO/metal/Si₃N₄ arrangement beating both pure stacks and matching what industry does. A better architecture — two metal layers reaching LSG 1.76 against a single-metal ceiling of 1.37. And a specified climate dependency: the two profiles share one candidate out of ten.

What it corrected. Six defects in the proposed weighting, one of which let the optimiser return 29% more silver than the incumbent at a perfect score. Two citations that cannot be traced. One measurement inconsistent with a model-independent limit, with six explanations eliminated. And five of its own conclusions, overturned by finer resolution or better sources and recorded rather than hidden.

What it cannot claim. Nothing has been deposited. The candidate rankings are hypotheses ordered by physics, suitable for directing experimental effort. Their principal weakness is known, quantified, and addressed by a specified experiment.

The recommendation. One XRD scan, one film, an afternoon — measuring the grain size that §5.7 identifies as the common cause of two separate limitations, testing the templating mechanism directly, and discriminating the microstructure hypotheses by a second independent route.

And two questions to Saint-Gobain: **which stack is the right baseline**, and **which transmittance target is real**.

A model that identifies the measurement capable of falsifying it is more useful to an experimental programme than one that does not. That is what this framework is for, and it is what it delivered.